Temporal Profile Analyzer

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Submission Documentation

The Temporal Profile Analyzer is an interactive Python program that processes a user's name and age to calculate an AI Era Readiness Score. The program removes unnecessary whitespace from the entered name, calculates its character length, and converts it to Title Case. It also validates the user's age using `isdigit()` before converting it from a string to an integer. Using the current calendar year obtained from the `datetime` module, the program calculates the user's projected age in 2045 and uses it together with the name length to calculate the AI Readiness Score. The result is displayed as a floating-point value rounded to two decimal places. As a bonus feature, the program repeats the formatted name according to the first digit of the user's age.

Python Source Code

```
"""
Program : Temporal Profile Analyzer
Purpose : Computes an AI Era Readiness Score from user metadata.
Author : Punit Kumar
Date : 08 August 2026
"""

from datetime import date


def numeric_age(age):
    return age.isdigit()


def temporal_profile_analyzer():
```

```
    """Here user have to write their full name"""
    full_name = input("Enter Your Full Name: ").strip()

    if not full_name:
        print("Please Enter Your Full Name")
        return

    # Age Input as numeric
    current_age = input("Enter Your Current Age: ")

    if not numeric_age(current_age):
        print("Please Enter a Valid Numeric Age")
        return

    # converting from str to int
    current_age = int(current_age)

    # Name Analysis
    formatted_name = full_name.title()
    name_length = len(full_name)

    # Finding current year using datetime library
    current_year = date.today().year

    # Projected Age in 2045
    age_in_2045 = current_age + (2045 - current_year)

    # AI Readiness Score
    ai_readiness_score = ((name_length * 10) + age_in_2045) / 2

    # Bonus - repeating the name based on first digit of age
    # age // 10 gives first digit for two digit age
    # for single digit age, treat it as 1 time
    first_digit = current_age // 10

    if first_digit == 0:
        first_digit = 1

    repeated_name = formatted_name * first_digit

    # Output
```

```
    print("\n========== Temporal Profile Analyzer ==========")
    print(f"Formatted Name : {formatted_name}")
    print(f"Identifier Byte-Count : {name_length}")
    print(f"Current Age : {current_age}")
    print(f"Current Year : {current_year}")
    print(f"Projected Age (2045) : {age_in_2045}")
    print(f"AI Readiness Score : {ai_readiness_score:.2f}")
    print(f"Repeated Name (Bonus) : {repeated_name}")
    print("===============================================")


temporal_profile_analyzer()
```

Program Output

Perfect Run

The following is a sample successful run using the worked example from the assignment, with the current year being 2026.

```
Enter Your Full Name: Scoooby-Dooo
Enter Your Current Age: 20

========== Temporal Profile Analyzer ==========
Formatted Name : Scoooby-Dooo
Identifier Byte-Count : 12
Current Age : 20
Current Year : 2026
Projected Age (2045) : 39
AI Readiness Score : 79.50
Repeated Name (Bonus) : Scoooby-DoooScoooby-Dooo
===============================================
```

Failed Run

The following sample demonstrates the handling of invalid non-numeric age input.

```
Enter Your Full Name: Scoooby-Dooo
Enter Your Current Age: twenty
Please Enter a Valid Numeric Age
```

Conclusion

The Temporal Profile Analyzer successfully demonstrates the use of Python data types, explicit type conversion, string processing, formatted output, and conditional validation. The program obtains the current year dynamically using the `datetime` module, projects the user's age to 2045, and calculates the AI Readiness Score according to the required formula. The program also handles invalid age input gracefully using `isdigit()` and includes the bonus string-repetition feature. Overall, the program follows the main computational requirements of the Temporal Profile Analyzer practical.